CLA-3 – Case Study

Machine Learning

Case study title	Iris Flower Species Classification Using kNN
Base paper	Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. Annals of Eugenics, 7(2), 179–188.
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1. Introduction and Problem Statement

Classifying different types of flowers correctly is important in many areas like agriculture and plant research. Doing this by hand takes a lot of time and needs someone who knows the plants well. Machine learning can help by looking at measurements of the flower and automatically saying which species it belongs to.

In this case study I am trying to answer this question: Can we use the k-Nearest Neighbors algorithm to correctly classify the three species of Iris flower (setosa, versicolor and virginica) just by using four measurements – sepal length, sepal width, petal length and petal width?

I chose this problem because the Iris dataset is small and clean, so it is good for understanding how a basic classification algorithm works in practice.

2. Base Paper Summary

The original paper for this dataset is written by R. A. Fisher in 1936. The title is “The use of multiple measurements in taxonomic problems.” In that paper Fisher used linear discriminant analysis to show that the three Iris species can be separated using the four measurements of sepal and petal.

He found that Iris setosa is very easy to separate from the other two, while versicolor and virginica have some overlap. The dataset later became very popular in machine learning. In my case study I am using the same data but I am applying the kNN algorithm instead of linear discriminant analysis to see how well a simple distance-based method works.

3. Method / Model Explanation

k-Nearest Neighbors (kNN) is a simple supervised learning algorithm. It does not really “train” a model in the usual sense. It just stores all the training data and when a new sample comes, it looks at the k closest points and takes a majority vote of their labels.

How it works step by step:

1. Keep the entire training set with labels.
2. Decide the value of k (I used k = 5) and choose a distance measure (I used Euclidean distance).
3. For a new flower, calculate the distance to every training sample.
4. Pick the k samples that are closest.
5. Look at the labels of those k samples and choose the class that appears most often. That becomes the prediction.

Main formula (Euclidean distance):

$$d(x_q, x_i) = \sqrt{\sum_{j=1}^{4} (x_{q,j} - x_{i,j})^2}$$

Here x_q is the new sample we want to classify and x_i is a training sample. There are 4 features, so we sum the squared differences of all four measurements and then take square root. After finding the distances, we simply take the majority class among the 5 nearest neighbours.

Small example:

Suppose we have a new flower with measurements [5.1, 3.5, 1.4, 0.2] and k = 3. If the three nearest training points all belong to setosa, the algorithm will predict setosa. If two of them are versicolor and one is setosa, it will predict versicolor.

4. Implementation

I used Python and the scikit-learn library. The main class I used is KNeighborsClassifier. The Iris dataset is already available inside scikit-learn, so I did not have to download any extra file. The code is uploaded on my GitHub repository.

The dataset has 150 samples in total. Each sample has 4 features and belongs to one of the three classes. There are exactly 50 samples of each species, so the data is balanced.

Settings I used:

- Split the data into 70% training (105 samples) and 30% testing (45 samples) with random_state = 42 and stratify = y so that the class ratio stays the same.
- k = 5 and Euclidean distance.
- Trained the model with fit() and made predictions with predict().
- Calculated accuracy, classification report and confusion matrix.

One problem I faced was a version mismatch between NumPy and scikit-learn in the environment I was using. After fixing the NumPy version the code ran without any other issues. The Iris data itself did not need any cleaning or scaling because all the features are already in similar ranges.

5. Results and Discussion

On the test set of 45 samples the model gave an accuracy of 97.78%. Only one sample was predicted wrongly.

Class	Precision	Recall	F1-Score	Support
setosa	1.00	1.00	1.00	15
versicolor	0.94	1.00	0.97	15
virginica	1.00	0.93	0.97	15
Accuracy			**0.98**	45

Table 1. Classification report (test set, k = 5)

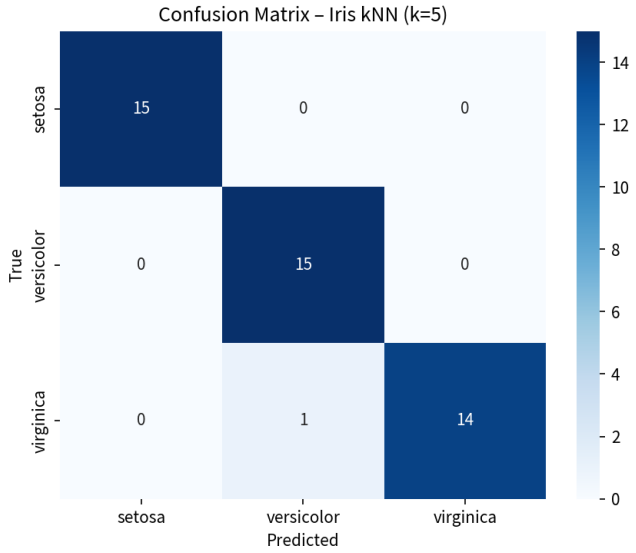

Figure 1. Confusion matrix of the kNN model

Correct prediction example: One test sample had features sepal length = 5.4, sepal width = 3.4, petal length = 1.7, petal width = 0.2. The true label was setosa and the model also predicted setosa. These values are typical for setosa, so the neighbours were all setosa and the prediction was correct.

Incorrect prediction example: One virginica sample with measurements [4.9, 2.5, 4.5, 1.7] was predicted as versicolor. This sample is close to the boundary between versicolor and virginica. Among its five nearest neighbours, more were versicolor, so the model made a wrong decision. This was the only mistake in the whole test set.

Overall the results are good. The four measurements are enough to separate the three species with high accuracy. The single error matches what Fisher also observed – versicolor and virginica sometimes overlap. Some limitations are that the dataset is very small and balanced, so the same accuracy may not appear on larger or noisy real-world data. Also, the performance of kNN depends a lot on the choice of k.

6. Conclusion

The question I asked in the beginning was whether kNN can correctly classify the three Iris species using only the four flower measurements. From the results, the answer is yes. With k = 5 the model reached 97.78% accuracy and made only one mistake on 45 test samples. Setosa was classified perfectly, and the only confusion was between versicolor and virginica. This shows that even a simple algorithm like kNN works well for this problem. At the same time, the single misclassification reminds us that the method can still make mistakes when samples lie near the decision boundary.

7. References

[1] R. A. Fisher, “The use of multiple measurements in taxonomic problems,” Annals of Eugenics, vol. 7, no. 2, pp. 179–188, 1936.

[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[3] E. Anderson, “The irises of the Gaspe Peninsula,” Bulletin of the American Iris Society, vol. 59, pp. 2–5, 1935.

[4] Scikit-learn, “Iris plants dataset,” https://scikit-learn.org/stable/datasets/toy_dataset.html#iris-plants-dataset (accessed Nov. 2026).

8. Declaration

I confirm that this case study is my own work and that all results reported were obtained by me.

Signature: _____

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